What is Economics?

Economics is a social science concerned with the production, distribution and consumption of goods and services and studies how individuals, businesses, governments and nations make choices on allocating resources to satisfy their wants and needs and tries to determine how these groups should organize and coordinate efforts to achieve maximum output.

Engineering Economics

Refers to the analysis and evaluation of monetary consequences by using the theories and principles of economics to engineering applications, designs and projects.

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Interest

Is the charge for borrowing of money or the amount paid to the borrowed capital.

Nominal Rate of Interest

Nominal Rate (NR) is the advertised rate and is the sum of the periodic rates in one-year period. Also known as Annualized Percentage Rate (APR).

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$$i = \frac{NR}{m}$$

$$n = Nm$$

Where:
m = number of time money earns interest in one year
N = number of years

Example: Excel Review Center

12 % compounded quarterly for 5 years

NR = 0.12 m = 4 N = 5

Mode of compounding:
Annually –> m = 1
Semiannually -> m = 2
Quarterly -> m = 4
Bimonthly -> m = 6
Monthly -> m = 12
Semimonthly -> m = 24
Daily -> m = 365

Effective Rate of Interest

Effective Rate (ER) is the annual rate of interest earned during a one-year period.

$$ER = (1+i)^m - 1$$

Where:
m = number of time money earns interest in one year

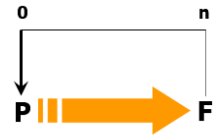
Simple Interest

This type of interest is computed only on the principle and is commonly used in variable rate consumer lending and in mortgage load where a borrower pay interest only on funds used. Usually this type of interest is used for a period less than one year. Excel Review Center

$$I = Pin$$

Where:
P = Principal
i = interest per period
n = number of periods

FUTURE AMOUNT, F



$$F = P(1+in)$$

Ordinary Simple Interest

Based on one Banker's year. Banker's year is a year that consists of 12 months, each having 30 days and amounts to a 360-day year.

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$$n = \frac{d}{360}$$

Where:
d = number of days

Exact Simple Interest

Based on the exact number of days in a given year.

Normal Year:

$$n = \frac{d}{365}$$

Leap Year:

$$n = \frac{d}{366}$$

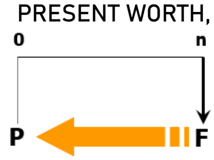
Compound Interest

This type of interest is computed on the principle as well as the interest already earned. The interest may be compounded monthly, quarterly, semiannually or annually, etc.

FUTURE AMOUNT, F

$$F = P(1+i)^n$$

PRESENT WORTH, P



$$P = \frac{F}{(1+i)^n}$$

Where:
i = interest per period
n = number of periods

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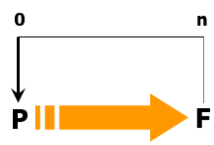
Continuously compounded

Unlike compound interest where the money earns interest quarterly or annually depending on how it was compounded, money invested in continuously compounded interest earns interest every second immediately starting from the time the money is invested.

FUTURE AMOUNT, F

$$F = Pe^{rt}$$



Where: Excel Review Center
P = Principal
e = Mathematical constant, Euler's number = 2.718...
r = rate of interest
t = period in years

Inflation

It is the rate at which the general level of prices for goods and services is rising and consequently, the purchasing power of currency is falling.

$$F = P\frac{(1+i)^n}{(1+f)^n}$$

Where:
P = Principal
i = interest rate
f = inflation rate

Discount

Refers to the difference between the future worth and the present worth.

The rate of discount, d is the discount on one unit of principal per unit time. Excel Review Center

$$d = 1 - \frac{1}{1+i}$$

$$i = \frac{d}{1-d}$$